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Structured Methods of New Product Development and Creativity Management: A Teaching Experience

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Using structured methods for managing business innovation can be an effective way to improve the ideation process. Teaching structured methods is a potent way to enhance the innovative capabilities of companies and to develop creative products for the marketplace. In this paper, structured models of innovation and product development are reviewed and an approach to them is presented, based on several MSc student projects. The implications for managers, such as the need to train employees in structured methods, and the implications for those who teach innovation management are discussed in detail.

Introduction

From time to time innovation is rediscovered as a prime enabler of growth (Kanter, 2006). A recent survey directed at 3,000 global executives conducted by BCG (2008) shows that innovation continues to represent a top strategic focus point for many companies: 63 per cent of the executives surveyed ranked this as being the first or at least among the top three strategic priorities compared with 67 per cent in 2007. Thus, innovation remains at the top of the corporate agenda. Innovation, as Drucker (2002) argued, is mostly the result of hard work rather than a product of genius; the most innovative business ideas are thus considered to be the result of a methodical search for new opportunities and ideas. This continual quest for breakthrough ideas for products, services and novel means to commercialize them has led some authors to proclaim that a new paradigm of innovation based on creativity is emerging (*Business Week*, 2005). Particular attention is nowadays being devoted to the topics of creativity and the use of structured methods and the manner in which they can foster improvements in the product development process.

In this paper, the experience of teaching structured methods for innovation and product development in a postgraduate

educational context is explored. Through the discussion of several case studies, based on projects developed by student teams, the authors seek to demonstrate not only the usefulness of systematic methods of product design, but also that these methods, when applied correctly, do not stifle creativity but enable more significant product concepts. The managerial implications, including the need to train employees in structured methods and creativity, are also analysed.

This paper is structured in four sections. The first presents a literature review covering topics such as creativity, structured methodologies and NPD (new product development) education. The second section deals with the methods used as well as the presentation of the case studies. A cross-case analysis and discussion is presented in the third section. Conclusions along with teaching and managerial implications are presented in the final section.

Literature Survey on Creativity, Structured Methods and NPD Teaching

The Role of Creativity

Creativity, defined as the ability to come up with novel ideas that promote innovation, is

not a new concept. In 1963, Levitt (1963) had already discussed the importance of creativity, although arguing that 'creativity, as an engine of innovation, is not enough, if the creative people do not take the responsibility for implementation, considering that ideas that are not implemented are useless as a source of innovation'. This raises a point that continues to be relevant today: companies need to have a clear understanding of how to organize their resources to enable more creativity in the innovation process and to achieve significant financial results.

Creativity is a complex phenomenon, involving four main aspects (Kryssanov, Tamaki & Kitamura, 1999): the creative process, the creative agent, the creative situation and the creative product. The creative process, beginning with the definition of the conceptual question or problem, is considered to be of the utmost importance (Nemeth, 1997). Innovation is considered to be the practical implementation or solution to the problem in question, in a novel or unconventional way. The implicit adoption of premises and old or habitual patterns of solving problems stifles creativity (Nemeth, 1997). New techniques encourage people to stretch their creativity and think beyond what is conventional. Using different techniques can be the best way to avoid preconceptions, and creativity may be achieved through unstructured brainstorming, or through the application of formal tools. Coyne, Clifford and Dye (2007) suggest that the middle path between these two extremes may be the best approach and recommend 'creating new boxes' for people to think within, as this prevents them from losing their target/reference point. To a certain extent this idea is reiterated by Ogle (2007), who considers that breakthrough creativity can be considered as a network of *idea-spaces*, connected by strong and weak ties. Within an idea-space there will be links and between spaces there will be ties. The stronger the tie, the greater is the degree of internal cohesion/coherence within the node. On the other hand, the weaker the tie is, the longer is the reach of the link. Ogle considers that breakthrough creativity is generally associated with the integration of *idea-spaces* with the greater reach or weaker ties of the network (see Figure 16).

While creativity is generally not considered as synonymous with intelligence, education and experience may enhance creativity (Kryssanov, Tamaki & Kitamura, 1999), and influence the *thinking style* developed by the individual. The organizational and physical environment of the place of work also influences creativity (Shapero, 1985; Amabile, 1996). Several authors agree that *creative think-*

ing or ideation is a systematic process that can be managed like any other process. Shapero (1985), for example, has proposed a process, consisting of several phases or steps; whereas Wallas (1976) proposed a model consisting of four main stages: preparation (problem perception or experience), incubation (a gestation period through unconscious mechanisms), illumination or insight (when the ideas for a solution appeared) and verification or evaluation (when ideas are tested and their validity and utility are assessed). Coyne, Clifford and Dye (2007) propose a fully orchestrated ideation process, in which the scope of thinking is focused through the use of small number of brainstorming sessions addressing a set of previously defined questions, which act as triggers for the idea generation process. This has been proven to be a very effective way of generating good or even exceptional ideas. De Bono (2005) proposed a method known as the *Six Hats*, and introduced the concept of *lateral thinking* (De Bono, 1970) to translate the application of logic to creative thinking. Other concepts such as *emotional design* have also been proposed (Norman, 2004).

Whatever creative process is used, it will be associated with two cognitive mechanisms (Kryssanov, Tamaki & Kitamura, 1999, Hsiao & Chou, 2004): a form of *divergent thinking*, which is the ability to generate original, distinct and elaborate ideas; and a form of *convergent thinking*, which is the ability to logically evaluate and find the best solution from a variety of feasible alternatives. A parallel between these two mechanisms and the four basic cognitive operations proposed by Stemple and Badke-Schaub (2002) can be drawn: the generation and exploration processes serve to widen the design space through the use of *divergent thinking* and the comparison and selection phases represent *convergent thinking*.

Although creative thinking by its nature is unpredictable, it is generally recognized that structured methods can bring several benefits to the product development process. In particular, it has been observed that if members of a project team are not only taught the specific methods, but also how to self-reflect in an integrated and balanced way, creative thinking can indeed be enhanced (Altier, 1988; Stemple and Badke-Schaub, 2002; West, 2002; Leenders, Van Engelen & Kratzer, 2007).

The creative process, which traditionally has been considered a process without any rules, is currently seen as something that can and should be managed like any other process. Independently of any interpretation of the meaning and working mechanisms involved in the creative process, several tech-

niques have been developed to foster creativity, such as brainmapping, the four quadrant model based on brain research, classical brainstorming, brainwriting and computer-aided brainstorming (Lynch, 1990; Holt, 1996; Ulrich & Eppinger, 2008). Brainwriting is a form of written brainstorming in which participants write their ideas on special forms or cards that are then circulated within the group. This gives people more time to think than in classical brainstorming where the ideas are expressed spontaneously. A number of software programs exist for computer-aided brainstorming.

Structured Methods of New Product Development

Today's business environment requires companies to develop new products in an effective, efficient and sustainable manner in order to guarantee their long-term survival. At the same time, modern NPD projects are very often complex and risky, and thus demand tight planning and coordination. This in turn requires the implementation of structured design methods (Leenders, Van Engelen & Kratzer, 2007). Structured design methods are also known as systematic design methods.

Successful products are usually developed through the collective efforts of individuals in NPD teams, composed of creative people generally seen as being non-conformist, and thus the management of their creativity is critical (Shapero, 1985; Nemeth, 1997; Leenders, Van Engelen & Kratzer, 2007). Leenders, Van Engelen & Kratzer (2007) studied the effect of using structured methods on the creative performance of NPD teams, taking a particular look at the pattern of communication among team members. They found that a systematic design approach 'need not hamper team-level creative achievement as long as systematic design is used with care'. According to Leenders et al., the approach requires the balanced use of the four principles which they argue characterize modern systematic design. Firstly, the principle of *hierarchical decomposition* needs to be applied (i.e., breaking the overall product down into semi-independent components). Then, the principle of *systematic variation*, which brings about variations on established themes through the reordering and recombination of existing elements. Thirdly, the *principle of satisficing* (i.e., finding alternatives that provide an acceptable solution, following a search over a reasonable period of time). Finally, the *principle of discursiveness* (where solutions are sought in a step-by-step approach versus the use of intuitive methods

in which the problem-solving process takes place unconsciously thus requiring undisturbed thinking).

Stemple and Badke-Schaub (2002) studied team communication and the thinking process in design and proposed a theory of the basic elements of design thinking, which is composed of four basic cognitive operations: generation, exploration, comparison and selection. In design space in general or in a specific problem space, Stemple and Badke-Schaub believe that the problem-solving process should aim to match a *goal space* (which is limited in terms of its flexibility) with the *solution space* (which is very large, since it represents many possible solutions in general). Solutions are generated and then are explored in the light of the goal space. Figure 1 illustrates the generic model of the design process and the underlying basic thinking operations.

By using the model depicted in Figure 1, Stemple and Badke-Schaub studied the interaction of three teams in a laboratory experiment and concluded that the teams spent two-thirds of their interaction time focusing on the *content* and one-third on the *group process*. Only 10 per cent of the *content* component was directed at activities related to the *goal space*, the remaining 90 per cent was focused on the *solution space*. The pattern of thinking found differed among the various teams: in two of the teams, the solution ideas were immediately evaluated before any analysis took place, whilst in the remaining teams the analysis was carried out before the evaluation. The latter approach yielded better qualitative results, although it required more time and effort. Thus, Stemple and Badke-Schaub concluded that, when teaching methods to structure the design process, it is important to convey the importance of continuous self-reflection in order to optimize the thinking process.

Several models of product development have been proposed and used in the past: from traditional craftwork to new models based on more dynamic and flexible methodologies (such as those reviewed by Hatchuel, Le Masson & Weil, 2005, and Ulrich & Eppinger, 2008). Early development methods were informal but, as the complexity of projects increased, more disciplined approaches were proposed (Fitzgerald, 1998). Even so, Griffin (1997) found that 38.5 per cent of American firms did not use a formal process for managing NPD. Griffin also concluded that the best performing firms are more likely to have implemented NPD processes with formal stages and gates, rather than functional and sequential approaches. In a study of 776 organizations, Fitzgerald (1998) observed that 60 per cent of the organizations did not use a

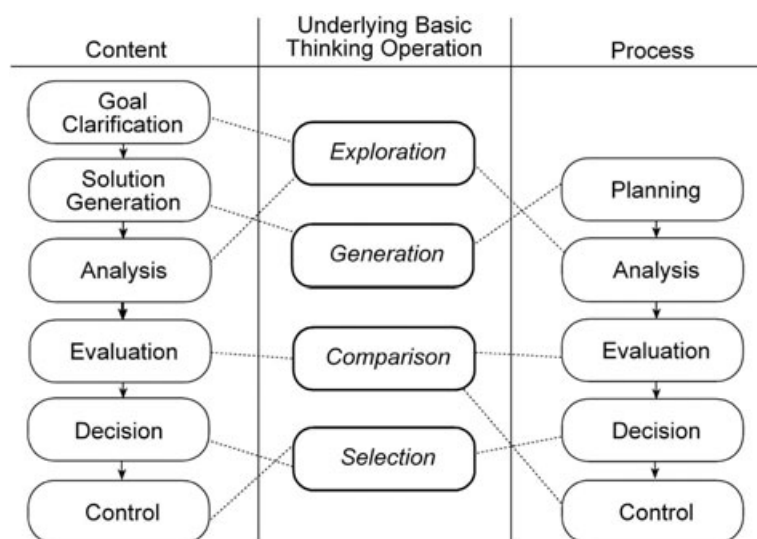


Figure 1. Generic Model of Design Process Thinking (adapted from Stemple & Badke-Schaub, 2002)

formal NPD process. Those that did use a formal process tended to regard it negatively, believing that the process led to inertia in the development process. On a more positive note, they perceived that formal processes facilitated project control and increased the transparency of the development process.

Today's product development processes are generally centred around a formal systematic design model, such as the model proposed by Pahl and Beitz (1996), which distinguishes three stages in the design process (functional, conceptual and embodiment design); or the stage gate process proposed by Cooper (1990); or competing engineering practices (Yazdani & Holmes, 1999). In the stage gate approach, the development process is generally carried out in a sequential manner and controlled by formal reviews between consecutive phases. The sequential models represent the traditional paradigm, as described by Cooper (1995) and the Ulrich and Eppinger model (2008). Recently, Cooper (2008) proposed an updated version of the stage gate model, which is flexible and can account for both the uncertainty in the external environment and the greater complexity of products. Hatchuel, Le Masson and Weil (2005) proposed a new management model for product development based on the notion of *design spaces* and supported by the C-K theory, where the authors try to include creative thinking and innovation in the design process.

Product development models provide a roadmap to transform an idea into a concept and a viable product. All the models aim to reduce development time and increase the

success rate of the innovation process by delivering more creative and innovative products. However, given the uncertainty associated with the development process, some authors (Thomke & Reinerstein, 1998) recommend that the process should be designed to be sufficiently flexible so that every contingency may be accounted for.

Teaching Product Development

Nowadays, it is assumed that new product design and business development can contribute to increasing a firm's competitiveness. This has led to renewed interest in researching and teaching new product development at post-graduate level. Courses on product design and innovation are offered by many universities around the world. Teaching experiences are being published by several authors, such as Lovejoy and Srinivasan (2002), Rosenau (2002) and Silvester et al. (2002) in a *Journal of Product Innovation Management* special issue on NPD education. More recently, Fixson (2008) reviewed product development teaching in North America. Further, a new approach to teaching innovation and NPD management was introduced at the 2008 International Product Development Management Conference (organized by EIASM) by several presentations on teaching experiences. Most of the courses described use the same philosophy; a combination of classroom and project work, where student teams have to identify an opportunity/market need and develop a product or service that satisfies it.

Conclusions on the Literature

The review of the literature suggests that the use of structured methodologies of product development is not as widespread as it needs to be if companies are to remain competitive in the future. The increasing number of courses offered by universities, on product development will contribute to raising the awareness to the importance of structured approaches to product innovation.

Methods

The aim of this study is to find out *whether, and to what extent, training on the deployment of structured methods for product design can lead to more creative and innovative product ideas*. To answer this question several projects were developed and analysed by students attending the course on 'Methods of Product Development', within the Masters programme in Industrial Design at the Faculty of Engineering of the University of Porto. The projects were developed over four successive academic years. Students with diverse backgrounds, such as mechanical engineers, architects, arts and graphics, industrial and interior designers, attend the programme. Teams of four to six students, who had training in engineering and design, were formed whenever possible. The aim was to mimic teams in an industrial setting. Although brainstorming was the main technique used by the different groups to generate ideas, other instruments to gather market data and information, such as questionnaires, interviews and observation were also used by the teams.

Four student projects (case studies 1–4) were selected for analysis and will be presented below. For each of these projects, which focused on developing a product concept, two student teams' results will be described. In the first project, the challenge provided was the development of a product concept that would satisfy a need that has already been partially met by products currently available in the market, albeit with some limitations. The second challenge, involved the creation of a new niche product, which is not yet available in the market. Thirdly, the students were asked to develop a product that meets the needs of a targeted customer group and the fourth project invited the students to develop a concept of a new child car seat that combined safety and entertainment features. The student teams were expected to produce drawings or virtual prototypes of their product concepts.

The entire exercise was meant to be a learning experience, since none of the students had

any previous experience and knowledge of product development methods. However, the students were encouraged to be creative, since innovativeness of the product concept was one of the evaluation criteria. All the groups received training on the product development model, proposed by Ulrich and Eppinger (2008), as the basic methodological framework that guided the design thinking process. However, they were free to interpret and implement the model in their own way. The following aspects which are likely to influence individual or team creativity were taken into consideration in the analysis:

- basic training of the team members
- coherent application of the methodology
- methods used to identify a real or latent need not satisfied by an existing product.

Case Study Description

Case Study 1: Pressure Cooker

The vision for this product was the development of a pressure cooker that would cook faster whilst preserving vitamins in the food. Student Team 1 began by conducting several consumer interviews to identify particular needs and attributes. Representatives of a pressure cooker manufacturer were interviewed, and applicable national standards were reviewed. Benchmarking of existing products was also carried out. On the basis of the information collected, the team derived a list of attributes and features to define the specification of the new product and support the concept development process. From the list of attributes the team drew an objectives tree illustrated in Figure 2, which framed the team's design space.

The team's focus on existing products led the students to develop a concept which would be an improvement on existing products, with new features such as a common lid for cookers with various volumes, a reduced number of components, interchangeability, easy storage, an innovative and strong visual impact, user-friendly interface simplification and easy maintenance (rubber sealing replacement). The case analysis showed that the team adopted a narrow view when defining their design space, ending up with a concept that in fact can be considered an incremental innovation on what was already in the market and the team had not developed a breakthrough product. The team made no attempt to define the specification of the product before designing the new concept. Since the dominant pre-existing design was adopted, the team felt no need to think of alternative designs and so

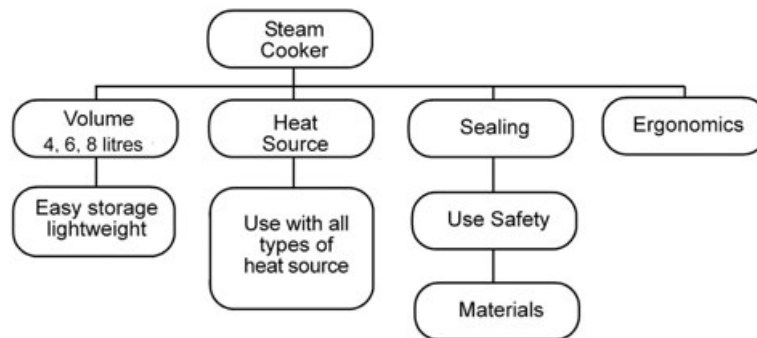


Figure 2. Objectives Tree with the Hierarchy of Attributes



Figure 3. Pressure Cooker with New Features following the Dominant Design

they based their work on introducing novel features to the current general concept, as illustrated in Figure 3. In practice what the team did was the development of a 'more efficient pressure cooker', thus attempting to make the product more appealing to potential users.

Student Team 2 followed a different path, placing emphasis on obtaining knowledge and information that supported the ideation process. On the basis of a customer survey and observation conducted by the team, they noticed that some users were afraid of cookers, since they thought they were unsafe. This triggered the team to think of the product in terms of its expected function, cooking healthy food by using pressurized steam to preserve its properties and reduce cooking time, rather than focusing solely on its current form.

Table 1 presents a sample of the information elicited from potential customers and the interpretation made by Student Team 2.

The analysis of customer needs revealed that safety ranked very highly and so any new concept should try to eliminate or at least reduce this perceived problem. Thus, they proposed a 'radical new concept': a pressure cooker that could be installed into a kitchen unit in the same way as a dishwasher or a fridge. The new concept of a 'domestic appliance' would need to be 100 per cent reliable in terms of safety and allow for the development of a family of new products with a much higher added value.

Case Study 2: Hip Car

This particular challenge was to design an innovative 'cool/fashionable' (i.e. 'hip' or 'in') concept for individual transportation (single-seater). The vehicle should have a good quality/cost ratio and should be appealing to teenagers and young adults between the ages of 16–28 and from families with medium to high incomes.

Team 1 based the needs identification process on a survey of the target market. A benchmarking of the products available in the market was also carried out. Group, individual and on-line questionnaires were used. The main target use of the vehicle was transportation to and from college or work, followed by leisure activities. Most of the interviewees travelled 20 kilometres per day, and most preferred to be powered by internal combustion engines, followed by hybrid vehicles. Cost, safety and comfort were the three main factors that influenced the purchasing decision; other valued attributes were exterior design, brand, environmental friendliness, luggage space and interior design. Benchmarking was conducted in relation to motorcycles and indirectly to city vehicles, which are currently marketed (generally two-seaters). In the concept development phase, several alternatives were considered in order to solve issues such as vehicle architecture, packaging, number of wheels.

Team 2 surveyed 116 persons by e-mail. A total of 18 major needs were identified, in par-

Table 1. Customer Statements Interpretation (Student Team 2)

Questions	Customer answer	Need interpretation
Common use	■ 'Fast cooking, quick meals'	■ Cooking with pressure cooker is faster than with a common pot
Qualities of the actual product	■ 'I can prepare meals quickly'	■ Cooking time
Defects of the actual product	■ 'Pots should be lighter'	■ Use of fewer components or lighter materials
	■ 'The lock system is complex'	■ Intuitive locking mechanism, easy to handle
	■ 'It's not possible to see the cooking in progress'	■ Cooking time and its program indicator
	■ 'Pressure cookers are too high'	■ Height/volume ratio adequate
Suggestions	■ 'I need more safety when it comes to pressure cooker use'	■ Lock and valve reliability
	■ 'I would like to see the water level inside the pot'	■ Total or partial transparency of the lid
	■ 'I would like to be able to profit from the steam'	■ Steam recovery system
	■ 'I would like to know what the temperature is inside'	■ Temperature measurement
Alternatives	■ 'Pressure cooker is not used to cook a fast meal; I can use a microwave which has additional functions'	■ A competitive pressure cooker should offer additional functions to consumers
	■ 'It should be autonomous thus not needing the stove'	■ Independent energy source

ticular: low consumption and cost, comfort, easy parking, safety, emissions reduction, luggage space, low maintenance costs, recreational needs, improved mobility, easy driving, allowing (facilitating) contact with the rest of the world, quality, versatility, attractive design, handling and easy fuel supply. On the basis of the attributes identified for the prospective product and information obtained through the questionnaires and benchmarking of rival products (see Figure 4), the team produced a detailed specification of the product in question. A quality function deployment (QFD) matrix was produced by the student team to identify the most important engineering metrics. The final concept was selected from a number of concepts developed by the team, as illustrated in Figure 5.

Team 2 considered that, in order for the new concept vehicle to be appealing to a young market, a special emphasis should be placed on technological innovations, in particular at the level of the powertrain, in addition to

having an *avant-garde* design. The final concept selected seems to embody such characteristics, as illustrated in Figure 6.

Case Study 3: A Product for Older People

Here the challenge was to identify an actual or latent need of older people (65 years or more) and then develop a product that was capable of satisfying that need. Since the students were not constrained to think in terms of a specific product, the idea space was much broader than in the previous cases. The teams conducted group and individual interviews to gather information about the targeted market. The needs that were identified were grouped into five categories: mobility, health, communication, domestic care and leisure. The data gathered during this phase was very important in the present case study so as to identify an opportunity and to direct the efforts of each team throughout the subsequent phases. Thus, one of the teams selected mobility as repre-



Figure 4. Benchmark of Competing Products

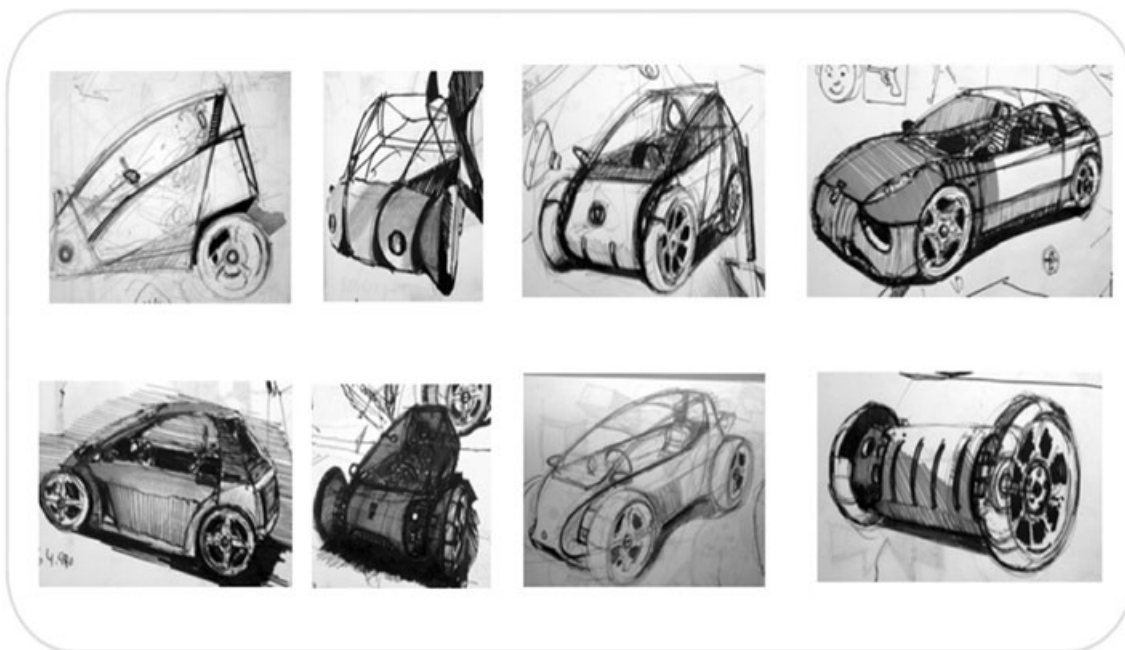


Figure 5. Concept Solution Space

sending the greatest concern for senior citizens, since a lack of mobility limits their autonomy in daily life (social and individual). Team 1 elected to develop a small vehicle for short distances, to make mobility more comfortable

and easier. The team followed a sequential product development methodology. A careful study of demographics and life expectancy was conducted to detect future trends among the ageing population, which will tend to

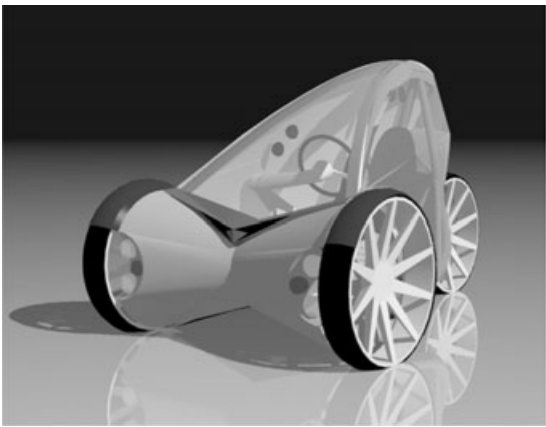


Figure 6. Hip Car Final Concept for Individual Transportation

increase in the future. A survey of a sample of 62 people with ages ranging from 64 to 81 was carried out. The main attributes identified are shown in Table 2.

A detailed specification was derived from the attributes required for the product. The concept generation phase is illustrated in Figure 7. Figure 8 illustrates the final concept: the 'R-walk'.

Team 2, on the other hand, proposed an automatic pill dispenser. Such a product is already available in some markets, notably in the United States, but the concept has the potential for further development. The concept solution space is illustrated in Figure 9 and the final selected concept in Figure 10.

Table 2. Mobility Main Attributes (Team 1)

Physical needs	Social needs	Legal needs	Environmental needs	Personal needs
Easy access to the vehicle Ergonomics Good visibility Low noise Comfortable Navigation aids Easy to park Weather protection	Low cost Attractive design Safety Reduced consumption Reduced maintenance	No driving licence	Non-polluting	Luggage space

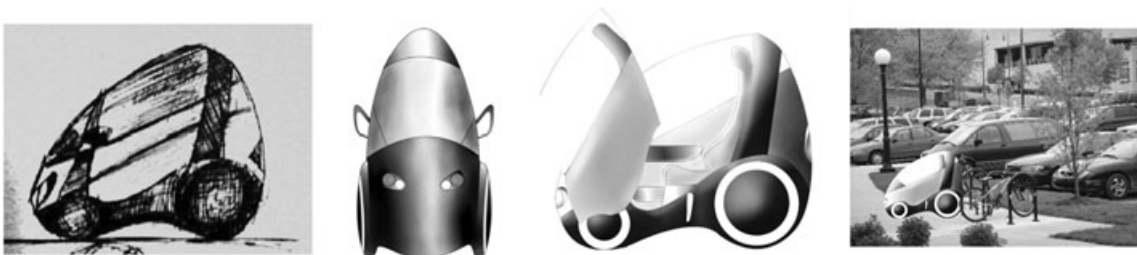


Figure 7. 'R-Walk' Concept Generation



Figure 8. 'R-Walk' Final Concept

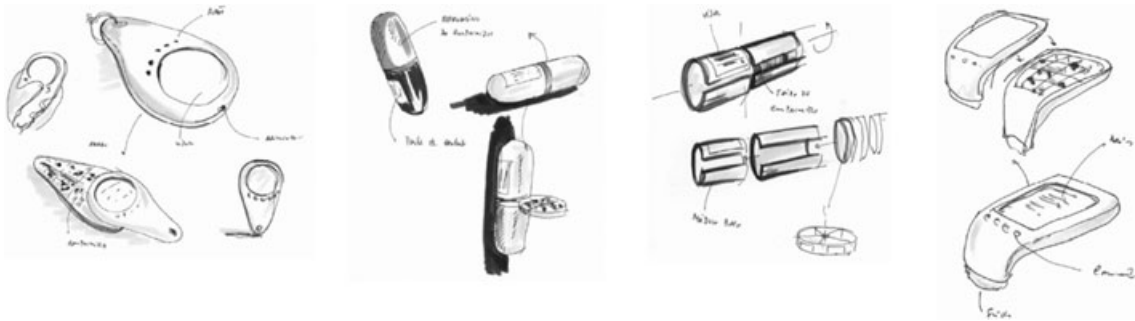


Figure 9. Several Proposed Concepts



Figure 10. MADC Pill Dispenser

Case Study 4: Entertainer Child Seat

This vision statement called for the development of a child seat, for children up to the age of 12, with built-in entertainment, to make trips more pleasant for both the children and their parents. The teams were asked to develop an innovative seat that had to comply with the current stringent safety standards.

Team 1 based the needs identification phase on a questionnaire, answered by 25 parents (a limited sample). When the information was processed, the team discovered that the questionnaire had insufficient open questions to allow the customers to express their needs, which in their opinion were not being met by the current products. The main features identified were related to safety, usability for children of various ages, cost, comfort and control

of the activities. Entertainment was not considered relevant and in most cases this was not even mentioned by parents. This may result from the fact that no such products are currently available on the market. The results of the questionnaire were compared with the features of competing products through a benchmarking exercise. Team 1 developed three concepts: a 'friend', a 'quiz-game' and playstation 'playcar' as illustrated in Figure 11.

The concepts proposed, by various members of the team were analysed and a screening process carried out by all members. On the basis of this democratic process, the team decided to develop a new concept, combining the best features of the concept 'friend' and 'quiz-game' as illustrated in Figure 12.

The team conceptualized the product as one that could be adapted to fit existing child

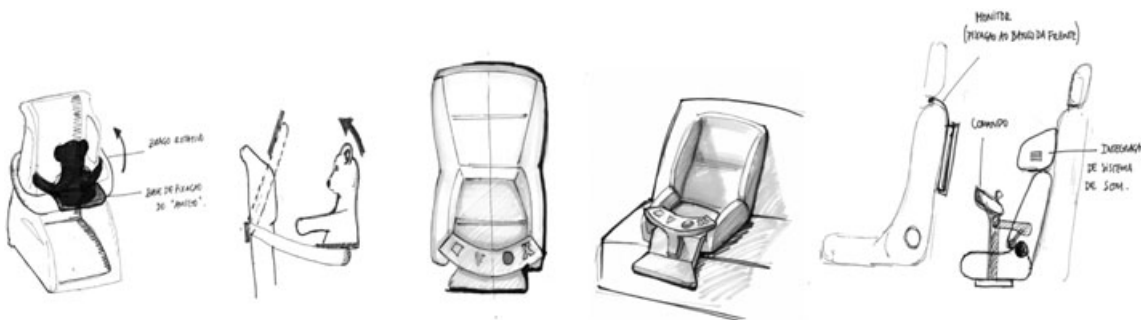


Figure 11. Concepts Proposed by Team 1



Figure 12. Final Concept



Figure 13. Children's Vision of the Seat

seats, which consumers generally considered adequate in terms of safety/comfort and cost. This would avoid competing directly with the main players in the market and creating a niche market, which in the opinion of the team continued to be underexplored.

Team 2 gathered ideas not only from parents but also from children (aged 8 to 12 years) as well as from those attending nursery school. It is interesting to mention the initiative of the team in getting feedback from the users of the seat themselves – children. Figure 13 illustrates the needs expressed by children and their vision of the product. One child stated that: 'If my parents owned a big American-type car I would love to watch DVD and play with a playstation', and another that 'when travelling I like to do things on my own'.

Team 2 analysed the information obtained from their various sources, including a benchmark of current products, to develop several concepts as illustrated in Figure 14. The final concept, illustrated in Figure 15, was selected by using the concept screening and scoring tools.

Analysis and Discussion

Cross-Case Analysis

Methodological approaches to new product development have been proposed over a

number of years by many authors and have been used extensively by companies, with varying degrees of success (Griffin, 1997). The dilemma facing many managers nowadays is the need to reconcile two apparently conflicting requirements: fostering creativity and maintaining discipline in the development process. This paper has looked at this dilemma in a comparative study of student projects, conducted during a course on new product development.

Table 3 shows a summary of the cross-case analysis of the student teams. It shows how the students integrated the feedback obtained from potential customers into their concepts and achieved varying levels of creativity. As the students were taught how to apply a structured NPD method like the stage gate, this provided a common framework for the project and they were also given several sub-assignments that acted as milestones for their project work.

During the early phase of the project work, the teams looked for opportunities for developing new products, through brainstorming sessions until they came up with an idea worth exploring in depth. The exception was the 'product for older people', which allowed the students to identify the benefit proposition, the stakeholders, target market and business goals.

During the subsequent phases of the project work (customer needs identification, specification, concept generation and selection), the students applied the structured methods in different ways. From our analysis, the path and outcome of the design thinking process followed by the different teams varied, and we see that the design spaces adopted by the teams were quite different. For example, in the case of the pressure cooker, Team 2 appeared to use the idea spaces of domestic appliances and their integration with the idea space of steam pressure cookers in addition to the perception of a lack of safety (justified or not), as external factors to enhance their creativity. Figure 16 shows the links between the group



Figure 14. Concept Solution Space of Team 2

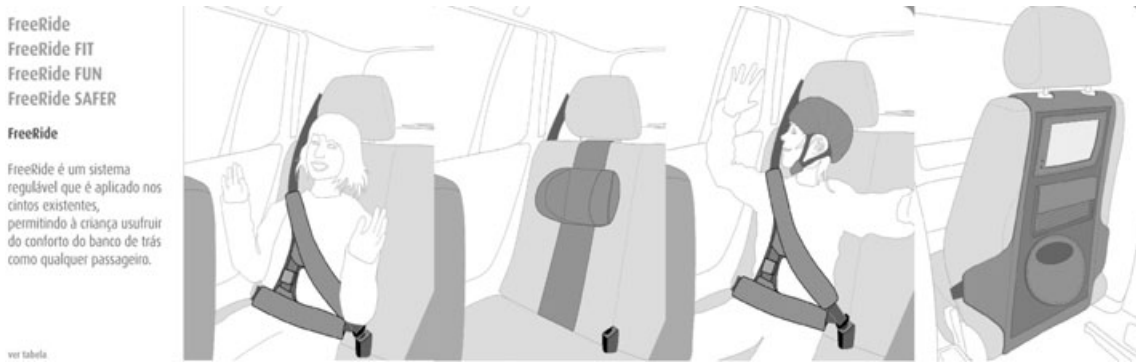


Figure 15. Final Concept

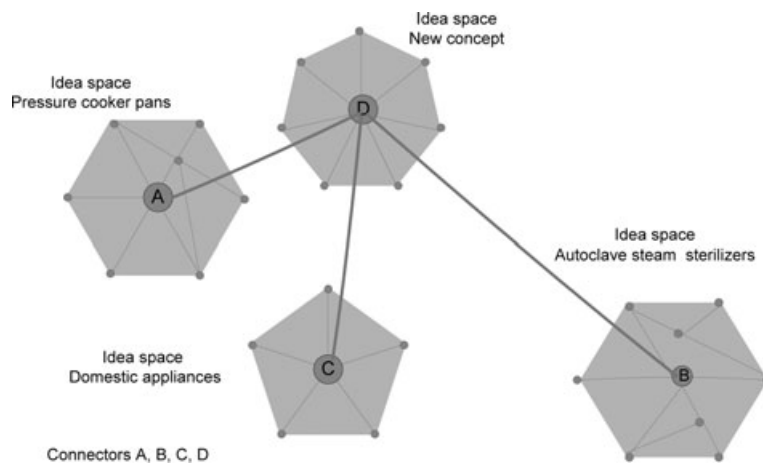


Figure 16. Network of Groups of Idea Spaces with Strong Internal Ties Linked by Weak Ties (adapted from Ogle, 2007) – Applied to the Pressure Cooker Case Study

of domestic appliances such as dishwashers or ovens and the pressure cookers.

A similar influence can be observed on both teams in the 'hip car' case study. In both case studies, the wider design space adopted has probably triggered the generation of more creative designs. In the case of the pressure cooker and hip car and, to a lesser extent, in the case of the child car seat, it has also been observed that the teams that spent more time in analysing the ideas, originated through a more in-depth identification of customer needs, benchmarking and the generation of

a wider range of concepts, were those that developed the most innovative concepts. This supports the findings of Stemple and Badke-Schaub (2002). In the case study on products for older people, the teams did not attempt to link their ideas to idea spaces that were remote from the products already available on the market, which directly or indirectly appears to have led them to be less creative on this assignment.

As previously mentioned, the design model followed was sequential and discursive in nature. Figure 17 depicts the sequence of

Table 3. Cross-Case analysis

Case	Project team	Approach	Outcome	Creativity level
Pressure Cooker	Team 1	Focused solely on traditional pressure cookers	Incremental innovation; increased modularity and functionality	Low
	Team 2	Looked into 'idea spaces' of other sectors	Integration of different 'ideas' led to a radical new concept of 'appliance', rather than a 'new pot'	High
Hip Car	Team 1	Followed a linear structured approach. Concept generation based on 'idea spaces' of motorcycles and city cars	Concept is based on features that satisfy user needs, but lacks emotional content	Low
	Team 2	Followed same approach as Team 1, drawing ideas from different idea spaces ('boxes')	Team selected the 'appeal' of the concept to youngsters, and the design proposed had greater emotional content	High
Product for the Elderly	Team 1	Identified real or latent needs of the elderly and translated this into a product through a structured approach; narrow focus on existing products	Concepts proposed can be considered to be incremental innovations of products already offered by the market	Low
	Team 2	Focused on the strategy of product development. Opted for a narrow approach in the identification of stakeholders		
Entertainment Child Seat	Team 1		Developed add-ons to current seats, without competing directly with main market players	Medium
	Team 2	In relation to Team 1, enlarged scope of needs identification, including the final users – children	Considering 'children' as end users gave a different focus to concept development	High

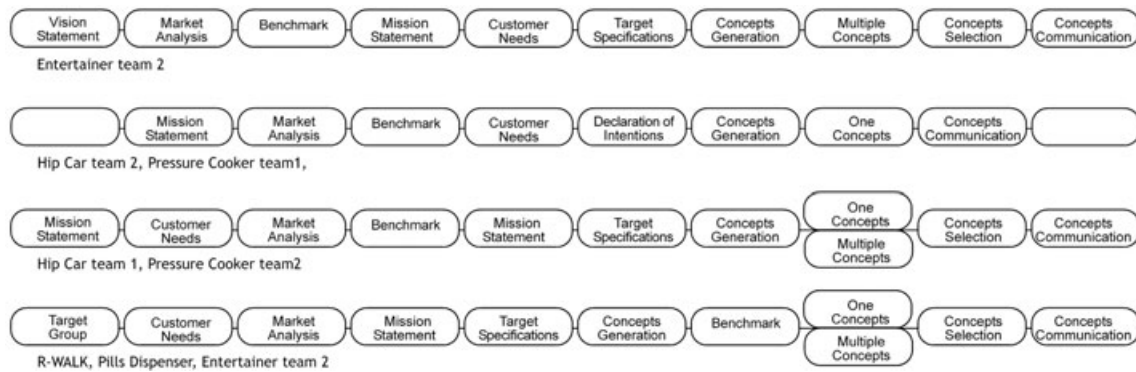


Figure 17. Product Development Path

activities performed by the various groups of students.

In the present study it is not clear what triggered the creativity of the different teams. Leybourne and Sadler-Smith (2006) analysed the role of intuition in project management and considered creativity to be prompted by external factors rather than internal factors. Ogle (2007), on the other hand, proposes a theory based on network dynamics of *idea spaces*: links between different idea spaces (illustrated in Figure 16) can, consciously or not, contribute to solving a problem and lead to the creative leaps necessary for a breakthrough.

The structured methodology that the students followed, illustrated in Figure 17, was generally seen as a framework that guided their thinking process. Interestingly, a number of the groups did not seem to be too constrained by the methodology, as can be observed from the variations that they used. It was also observed that, within the different phases, the emphasis and paths followed were not always identical. For example, some groups (pressure cooker and hip car) analysed the data gathered from the interpretation of the customer statements in greater detail. The other groups seemed to spend more time on the development of the concepts, without being too constrained by the boundaries imposed by the attributes identified in the 'needs identification' phase. It was also interesting to observe that groups who came from an engineering background tended to adopt a more technical approach, placing more emphasis on the technical solutions and defining the architecture. From the cross-case data available, it is not obvious whether the teams that followed the structured process more rigidly were more or less innovative than the others, but teams that presented solutions revealing a higher level of creativity had clearly thought deeply about redefining their concepts. Creativity in the context of product

development differs in nature from artistic creativity, which is based on the artist's ability to express and communicate. With products, creativity is the identification of needs; the communication of these through drawings and models; and the application of technology to achieve a solution. As observed by Akgun, Lynn and Yilmaz (2006), a key ability in new product development is to interpret information in different ways, often information which is apparently trivial and obvious. The ability to conceptualize can only be achieved effectively if an individual is able to switch to a different level of thinking and generate new concepts (Brock, 2006). In this thinking process, a human touch is essential to provide a product with a personality and an image.

Implications for Teaching

Our experience of teaching demonstrates that teamwork, coupled with the appropriate cognitive skills and customer feedback and aided by systematic design methodologies, can foster the creative performance of NPD teams (as has been reported by Leenders, Van Engelen & Kratzer, 2007). As Kelley rightly states 'Innovation is definitely not self-starting or self-perpetuating. People make it happen through their imagination, willpower, and perseverance; and whether you are a team member, a group leader, or an executive, your only real path to innovation is through people' (Kelley and Littman, 2006, p. 6). That is, the innovation process needs to be leveraged using structured approaches. The student case study projects described demonstrate that teams without previous experience in product development methods can, if duly trained, potentially achieve good results. As summarized by one of the teams, 'in spite of the difficulties inherent in the development of a complex product such as a vehicle, the inexpe-

rience of the group, an absence of knowledge of the method and a lack of resources in addition to time constraints... we felt that the application of a structured methodology was very important so as to foster knowledge sharing among members and establish a continuous link with the project goal and targeted market'. This demonstrates the importance of teaching structured methods for product development at universities. Training on structured methods for product design and ideation, as a means to support creativity while simultaneously maintaining control of the development process, is also relevant to companies. However, applying the findings of the teaching and learning research described in this paper to industry should be undertaken with caution given the different educational context.

Conclusions

University teaching on structured methods for new product development, and the use of tools and techniques to enhance creativity, will undoubtedly improve the skills of students who will in the future become project team members in industrial settings. In this way, innovative approaches to teaching and learning will contribute to product innovation in companies.

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